# RouteWise AI: Shipment Route Intelligence System

**Optimizing Multi-Objective Global Freight Routing Under Uncertainty** | Aiqon Take-Home Assignment

## 1. Executive Overview & Problem Definition

Global container shipping decisions require navigating competing, non-linear objectives under operational uncertainty. A route offering low direct freight rates frequently passes through bottleneck choke points (e.g., the Suez Canal, Bab-el-Mandeb) or congested transshipment hubs (Singapore, Rotterdam), multiplying delay variance. Conversely, express multimodal routing provides high speed and schedule reliability at prohibitive cost premiums. **RouteWise AI** formulates shipment routing as a constrained multi-criteria decision analysis (MCDA) optimization problem paired with an explainable AI layer. Rather than treating route selection as an opaque black box, the system computes transparent, reproducible trade-off vectors and grounds an LLM assistant strictly in deterministic optimization outcomes.

## 2. Mathematical Formulation & Decision Architecture

Candidate routes are evaluated across three normalized core dimensions: Landed Cost (C), Transit Time (T), and Composite Risk (R). To ensure fair comparison across diverse trade corridors, attributes are normalized dynamically relative to each shipment's candidate pool.

### Normalization & Composite Scoring:

```
C_norm = (Cost - Cost_min) / max(Cost_max - Cost_min, ε)
T_norm = (Time - Time_min) / max(Time_max - Time_min, ε)
R_comp = 0.40 * P(delay) + 0.30 * (Congestion/100) + 0.30 * (Geopolitical/100)
Score_total = w_c * C_norm + w_t * T_norm + w_r * R_norm (where w_c + w_t + w_r = 1.0)
```

**Lower score indicates superior multi-objective performance.** Unlike naive rank-averaging, this continuous formulation maintains proportionality: an option that is only 2% more expensive but 40% faster receives an appropriately weighted advantage.

## 3. System Architecture: Decoupled Optimization vs. Generative AI

A fundamental design principle of RouteWise AI is the **strict separation of mathematical decision-making from natural language generation**. Large Language Models (LLMs) are notorious for stochastic arithmetic errors, boundary hallucinations, and inconsistent numerical ranking. Consequently, the system operates in two decoupled tiers:

| Component Layer | Technology / Implementation | Operational Responsibility |
|---|---|---|
| **Deterministic Optimizer** | Python, NumPy, pandas | Hard constraint enforcement, Pareto dominance calculation, normalization, continuous score synthesis. |
| **Explainability Engine** | Rule-Based Engine + Gemini LLM | Synthesizes audit-ready natural-language rationale, explains why runner-up lost, answers interactive what-if queries. |
| **Interactive Dashboard** | Streamlit, Plotly | Interactive parameter tuning, trade-off radar / scatter visualizations, scenario sensitivity analysis. |

## 4. Constraint Handling, Pareto Dominance & Robustness

Real-world logistics systems encounter tight customer service level agreements (SLAs) and strict budget caps. RouteWise AI addresses these through deterministic mechanisms:

**1. Hard Constraint Invalidation:** Routes violating max budget (C_i > C_max), max transit days (T_i > T_max), or max risk thresholds are immediately quarantined with explicit diagnostic failure messages (e.g., *'Exceeds max transit time by 4.5 days'*). They are disqualified from winning unless *all* candidates are infeasible.
**2. Graceful Infeasible Recovery:** If an operator specifies an impossible constraint set, the engine flags all_infeasible = True and returns a transparent 'best-effort' recommendation ranked by minimal penalty distance rather than throwing an unhandled exception.
**3. Pareto Dominance Filtering:** Route A strictly dominates Route B (A ■ B) if C_A ≤ C_B, T_A ≤ T_B, and R_A ≤ R_B with at least one strict inequality. Dominated routes are flagged for logistics planners, providing mathematically verifiable justification for early route pruning.
**4. Zero-Division & Identical Metric Safeguard:** An ε = 10■■ threshold guarantees numerical stability when all route options in a corridor share identical attributes.

## 5. Empirical Evaluation & Sensitivity Benchmark

To test system behavior across diverse global freight dynamics, a synthetic benchmark dataset of 30 shipments across major corridors (Trans-Pacific, Asia-Europe, South Asia-Europe, Intra-Med) with 4 route archetypes each (120 options total) was generated. Each corridor includes: (1) Economic Standard Ocean, (2) Express Direct Ocean, (3) Resilient Cape Diversion, and (4) Multimodal Sea-Air Express.

| Scenario / Profile | Weight Profile (C/T/R) | Dominant Winner Archetype | Observed Operational Dynamics |
|---|---|---|---|
| **Balanced Baseline** | 40% / 30% / 30% | Express Direct Ocean | Balances moderate cost premium with 25% transit time reduction and low congestion. |
| **Cost-Focused** | 70% / 15% / 15% | Economic Standard Ocean | Captures lowest freight rate; accepts 7-10 extra days transit and higher port delay risk. |
| **Time-Critical** | 15% / 70% / 15% | Multimodal Sea-Air Express | Drastically cuts transit from 28d to 8d; incurs 2.2x cost acceptable for critical JIT components. |
| **Risk-Averse** | 15% / 15% / 70% | Resilient Cape Diversion | Bypasses geopolitical conflict zones (Red Sea); lower delay probability despite longer nautical path. |

***Key Sensitivity Finding:*** *Shifting from a Balanced to a Risk-Averse profile dynamically flips the optimal routing from Suez/Red Sea transshipment to the Cape of Good Hope corridor without human intervention, proving the engine's sensitivity to real-world maritime disruption scenarios.*

## 6. Explainability Framework & Grounded AI Verification

A critical evaluation metric in logistics decision support is auditability: operators will not trust recommendations they cannot explain to customers. RouteWise AI implements a dual-mode explainability architecture:

• **Grounded Deterministic Engine (Zero-Shot Fallback):** Translates exact delta vectors into executive explanations without network or API dependencies. It explicitly reports percentage cost differences, transit delta days, risk factors, and reasons why alternatives were rejected.
• **Grounded LLM Prompting:** When an API key is present, the verified mathematical result (JSON payload containing scores, trade-off deltas, and constraints) is injected into an LLM prompt instructed to act as an executive logistics advisor. Hallucination is strictly bounded because the LLM is prohibited from inventing values.

## 7. Practical System Limitations & Production Roadmap

While RouteWise AI provides robust multi-objective decision support, real-world deployment requires addressing key logistics realities:

| Limitation | Root Cause in Practice | Production Mitigation Strategy |
|---|---|---|
| **Static Risk Indices** | Synthetic congestion and geopolitical scores represent point-in-time snapshots. | Integrate live AIS satellite telemetry and port dwell-time APIs (e.g., MarineTraffic, Spire). |
| **Non-Linear Demurrage** | Delay costs are rarely linear; exceeding free-time windows causes step-function detention spikes. | Implement piecewise linear demurrage penalty functions conditioned on bill-of-lading terms. |
| **Deterministic vs. Stochastic** | Assumes fixed transit time estimates despite weather and labor strike volatility. | Upgrade scoring engine with Monte Carlo simulation and stochastic chance-constrained programming. |
| **Carbon / Scope 3 ESG** | Emissions currently treated as an informative attribute rather than an optimization penalty. | Add carbon cap constraints (IMO CII regulations) directly into the multi-objective Pareto solver. |

## 8. Conclusion & Deliverables Summary

RouteWise AI demonstrates that combining deterministic optimization algorithms with grounded LLM explanation layers yields a reliable, practical, and explainable decision-support tool for global logistics planners. All deliverables—including clean source code, 100% passing test suite, interactive Streamlit dashboard, and reproducible synthetic dataset—are packaged for evaluation in the GitHub repository.